

GRADE 7 · READING INFORMATIONAL TEXT · RI.7.4

# Word Meaning and Tone in Informational Text

Determine the meaning of words and phrases in an informational text about microplastics using context clues, and analyze how specific word choices shape the meaning and tone of the passage.

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Lesson Map · Passage · Guided · Partner · Independent Practice · Small Group · Answer Key

# What You Will Learn Today

## Learning Goals

Understand RI.7.4: Determine the meaning of words and phrases in an informational text ab...

Read an original passage closely

Support every answer with text evidence

Work through guided, partner, and independent practice

## How the Lesson Runs

Warm-up and passage

Guided practice together

Partner discussion

Independent + small-group work

Exit ticket and answer key

# Unpacking the Standard

Context clues help define unfamiliar words using nearby sentences. Word choice also shapes tone — the author's attitude toward the subject. A writer describing the same fact with "leaked" versus "seeped" creates a different feeling, even though both describe liquid escaping.

# Modeled Example

"The chemical leached slowly into the groundwater." What does "leached" likely mean, using context?

Drained or seeped out gradually — the clue "slowly into the groundwater" suggests a gradual movement of a substance into another material.

Compare "the factory released waste" to "the factory dumped waste." How does word choice change tone?

"Dumped" carries a more negative, careless tone than the neutral "released," even though both describe the same basic action.

# The Pieces We Cannot See

1. A single fleece jacket can shed over 100,000 microscopic plastic fibers in one wash cycle. These fibers, each thinner than a human hair, slip past most washing machine filters and travel with wastewater to treatment plants, which were not designed to catch particles this small.
2. Scientists call these fragments microplastics — plastic pieces smaller than five millimeters, roughly the size of a sesame seed or smaller. Some start large and break down over years, worn away by sunlight and waves. Others, like the fibers from synthetic clothing, begin small and never grow any larger.
3. Microplastics have now been detected in locations that once seemed impossibly remote: Antarctic sea ice, deep ocean trenches, and rainwater collected above mountain peaks far from any city. Researchers have also found them inside the tissue of fish, shellfish, and, increasingly, inside human blood and lung samples.
4. The health effects of this exposure remain an active area of research, and scientists are careful to avoid overstating conclusions that the evidence does not yet fully support. Some laboratory studies on animals suggest inflammation and cellular stress from high concentrations, but researchers caution that these doses often exceed typical human exposure.

# The Pieces We Cannot See, continued

5. What is well established is the scale of the problem: an estimated eight million metric tons of plastic enter the ocean each year, and once there, plastic does not readily biodegrade. Instead, it fragments slowly, meaning today's waste will keep breaking into smaller pieces for decades, even centuries.

6. Some manufacturers have begun installing filters on washing machines designed to trap fibers before they reach wastewater systems. A handful of countries have passed legislation requiring these filters in new machines, though adoption remains limited worldwide.

7. For researchers like Dr. Priya Chandrasekaran, who has spent a decade studying microplastic accumulation in coastal sediment, the challenge is not proving the problem exists — that evidence is now abundant — but determining, precisely, what long-term exposure means for both ocean ecosystems and human health.

# Warm-up · Before You Read

1. What does "shed" mean as used in paragraph 1 ("a single fleece jacket can shed...fibers")?
2. What context clue in paragraph 1 helps define "shed"?

# Guided Practice

3. What does "microscopic" mean, based on its use alongside "each thinner than a human hair"?

4. What does "fragments" mean as used in paragraph 2?

# Guided Practice

5. What tone is created by the phrase "impossibly remote" in paragraph 3?

6. How would the tone change if "impossibly remote" were replaced with "far away"? Explain.

# Partner Discussion

7. What does "overstating" mean in paragraph 4 ("avoid overstating conclusions")?

8. What tone do the words "careful" and "caution" create in paragraph 4?

# Partner Discussion

9. Why might the author choose careful, hedged language in paragraph 4 instead of more dramatic language?

# Independent Practice

10. What does "well established" mean in paragraph 5, and how does it contrast with paragraph 4's tone?

11. What does "biodegrade" mean, based on context in paragraph 5?

# Independent Practice

12. What effect does the phrase "for decades, even centuries" have on the passage's tone in paragraph 5?

13. What does "adoption" mean in paragraph 6 ("adoption remains limited worldwide")?

# Independent Practice

14. How does the tone of paragraph 6 differ from the tone of paragraph 3?

15. What does "abundant" mean in paragraph 7 ("that evidence is now abundant")?

# Small-Group Work

16. How does the final sentence of paragraph 7 shape the reader's overall impression of the article's tone?

17. Compare the tone of paragraph 1 to the tone of paragraph 4. How do they differ, and why?

# Small-Group Work

18. Find one word in the passage whose connotation (feeling) is more negative than a neutral synonym would be, and explain your choice.

19. How does the author's word choice throughout the passage balance concern with scientific responsibility?

# Exit Ticket

20. Choose one word from the passage and explain how its specific meaning, not just a general synonym, shapes your understanding of a sentence.

# Answer Key (1 of 4)

1. To release or lose small pieces — context clue: the fibers come off the jacket during washing and travel elsewhere.
2. "Over 100,000 microscopic plastic fibers in one wash cycle" and "slip past... filters" suggest fibers are coming loose from the jacket.
3. Extremely small, too small to see clearly without magnification — the comparison to a hair's thinness supports this meaning.
4. Small broken pieces — context clue: paragraph 2 explains some plastic "start[s] large and break[s] down over years," creating pieces.
5. A tone of alarm or surprise, emphasizing how unexpectedly widespread microplastic contamination has become.

# Answer Key (2 of 4)

6. The tone would become more neutral and less dramatic; "impossibly remote" emphasizes surprise and difficulty, while "far away" is a plain, unemotional description.

7. Exaggerating or claiming more than the evidence supports — the surrounding text explains that "the evidence does not yet fully support" strong conclusions.

8. A careful, measured, responsible tone, showing scientists are being precise rather than alarmist about uncertain findings.

9. To accurately represent scientific uncertainty and avoid misleading readers about health effects that are not yet fully proven, maintaining credibility.

10. Well established means proven or widely accepted as fact; it contrasts with paragraph 4's cautious, uncertain tone about health effects by presenting the scale of ocean plastic pollution as a settled, confirmed fact.

# Answer Key (3 of 4)

11. To break down naturally over time — context clue: "plastic does not readily biodegrade. Instead, it fragments slowly," contrasting biodegrading with slow fragmenting.

12. It creates a tone of long-term concern, emphasizing the lasting, seemingly permanent nature of the problem.

13. The act of countries or companies choosing to use something (here, washing machine filters) — context clue: it follows a description of some countries passing legislation requiring the filters.

14. Paragraph 6 has a more hopeful, solution-focused tone (describing filters and legislation), while paragraph 3 has an alarming tone (describing contamination in remote, unexpected places).

15. Plentiful or more than enough — context clue: it follows the claim that proving the problem exists is no longer the challenge, implying there is now a lot of supporting evidence.

# Answer Key (4 of 4)

16. It creates a tone of ongoing, serious scientific inquiry rather than alarm or certainty, framing the real challenge as understanding long-term effects, not doubting the problem's existence.

17. Paragraph 1 has a more startling, attention-grabbing tone (the striking 100,000-fiber statistic), while paragraph 4 has a cautious, restrained tone, reflecting the difference between describing a clear fact and discussing uncertain health research.

18. Answers vary; for example, "contamination" (implied throughout) carries a more negative connotation than a neutral word like "presence," emphasizing harm.

19. The author uses striking facts and words like "impossibly remote" to convey the scale of the problem, while also using cautious language like "careful," "caution," and "avoid overstating" when discussing uncertain health effects, balancing urgency with accuracy.

20. Answers vary; require a specific word, the context clue used to define it, and an explanation of how that precise meaning affects the sentence's impact.

# Exit Ticket

## Answer on a sticky note

Answer the exit-ticket question from today's lesson.

Cite at least one piece of evidence from the passage.

## Remember

Name your claim or answer

Quote or cite the evidence

Explain how it supports your answer